

Матричные производные

$f(x)$ куда откуда x	$\mathbb{R}$ скаляр	$\mathbb{R}^n$ вектор	матрица
скаляр $\mathbb{R}$	① $f'(x) \cdot dx$ $[1 \times 1] [1 \times 1]$	④ $\int dx$ $[m \times 1] [1 \times 1]$	
вектор $\mathbb{R}^n$	② $\nabla^T f \cdot dx$ $[1 \times n] [n \times 1]$	③ $\int dx$ $[m \times n] [n \times 1]$	
матрица			

Учитательские фокусы:

$$S$$

$$S^T = S$$

$$[1 \times 1]$$

Полезные св-ва:

$$(const)' = 0$$

$$dA = 0$$

$$(d \cdot f)' = d \cdot f'$$

$$d(\lambda \cdot X) = \lambda \cdot dX$$

$$(f \cdot g)' = f' \cdot g + f \cdot g'$$

$$d(xy) = dx \cdot y + x \cdot dy$$

$$\neq d(yx)$$

$$d(X^T) = (dX)^T$$

① Дифференциалы и производные

① $f(x): \mathbb{R} \rightarrow \mathbb{R}$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) - f(x) = \underset{\substack{\uparrow \\ f'(x)}}{A \cdot h} + \underbrace{O(h)}_{\text{мелкая функция}}$$

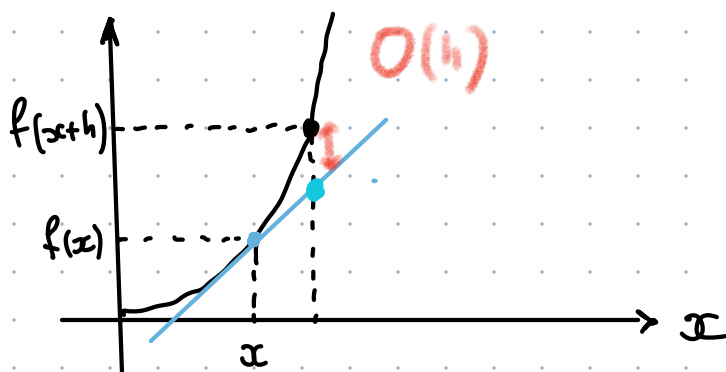
$$f'_t(t) \\ \frac{\partial f(t)}{\partial t}$$

$$f(x) = x^2$$

$$f'(x) = 2x$$

$$df(x) = f'(x) \cdot dx$$

$$df(x) = 2x \cdot dx$$



② $f(x_1, \dots, x_n): \mathbb{R}^n \rightarrow \mathbb{R}$

$$df(x) = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \dots + \frac{\partial f}{\partial x_n} dx_n =$$

$$= \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) \cdot \begin{pmatrix} dx_1 \\ \vdots \\ dx_n \end{pmatrix} = \langle \nabla_x f, dx \rangle = \nabla_x f^T \cdot dx$$

Упражнение

$$f(x) = a^T x$$

$$a, x \in \mathbb{R}^n$$

$$a = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \quad x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

$$\nabla_x f = ?$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(x) = (a_1, \dots, a_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = a_1 x_1 + \dots + a_n x_n$$

$$\frac{\partial f(x)}{\partial x_i} = a_i \quad \nabla_x f = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a$$

$$d(a^T x) = a^T dx \quad \nabla_x f = a$$

$$\nabla_x^T f \cdot dx$$

Упражнение

$$f(x) = x^T A x \quad x \in \mathbb{R}^n \quad A \in \mathbb{R}^{n \times n}$$

$$\begin{matrix} [1 \times 1] & [1 \times n] & [n \times 1] \\ & [n \times n] & \end{matrix}$$

$$\nabla_x f = ? \quad \nabla_x^2 f = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots \\ \vdots & & \end{pmatrix}$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$f(x) = (x_1, \dots, x_n) \cdot \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{i=1}^n \sum_{j=1}^n x_i x_j a_{ij}$$

$$\frac{\partial f}{\partial x_i} = ?$$

$$x^T A x$$

$$d(x^T A x) = d x^T \cdot A x + x^T \cdot d(A x) =$$

$$\begin{matrix} [1 \times n] & [n \times n] & [n \times 1] \\ [1 \times 1] & & \end{matrix}$$

$$d x^T \cdot A x$$

записано

$$x^T A dx$$

уже так

$$\nabla^T f \cdot dx$$

$$(AB)^T = B^T A^T$$

$$(d x^T A x)^T$$

$$x^T A^T dx$$

$$\nabla^T f$$

$$= x^T A^T dx + x^T A dx = x^T (A^T + A) dx$$

$$\nabla f = (A^T + A) x$$

$$\nabla_x^2 f(x) = H = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \dots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \dots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}$$

матрица
Гессе

III. проверка

$$H^T = H$$

$$g(x) = (A + A^T)x$$

$[n \times 1] \quad [n \times n] \quad [n \times 1]$

$$y = \nabla_x^2 f$$

$$dg = d(A + A^T)x = (A + A^T)dx$$

$$\textcircled{3} g: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$df = y \cdot dx$$

$$\begin{pmatrix} g_1(x_1, \dots, x_n) \\ g_2(x_1, \dots, x_n) \\ \dots \\ g_m(x_1, \dots, x_n) \end{pmatrix}$$

$$\begin{pmatrix} dg_1 \\ dg_2 \\ \vdots \\ dg_m \end{pmatrix}$$

$$[m \times 1]$$

$$= \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \dots & \frac{\partial g_1}{\partial x_n} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \dots & \frac{\partial g_2}{\partial x_n} \\ \dots & \dots & \dots & \dots \\ \frac{\partial g_m}{\partial x_1} & \frac{\partial g_m}{\partial x_2} & \dots & \frac{\partial g_m}{\partial x_n} \end{pmatrix}$$

$$[m \times n]$$

$$\begin{pmatrix} dx_1 \\ dx_2 \\ \vdots \\ dx_n \end{pmatrix}$$

$$[n \times 1]$$

$$f(x, y) = \begin{pmatrix} (x+y)^2 \\ \cos x \\ e^x \end{pmatrix}$$

$$df = \begin{pmatrix} d(x+y)^2 \\ d\cos x \\ de^x \end{pmatrix} = \begin{pmatrix} 2(x+y)dx + 2(x+y)dy \\ -\sin x \cdot dx + 0 \cdot dy \\ e^x dx + 0 \cdot dy \end{pmatrix} =$$

$$= \begin{pmatrix} 2(x+y) & 2(x+y) \\ -\sin x & 0 \\ e^x & 0 \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}$$

④ $f: \mathbb{R} \rightarrow \mathbb{R}^n$

$$x \mapsto \begin{pmatrix} f_1(x) \\ f_2(x) \\ \vdots \\ f_m(x) \end{pmatrix}$$

$$df = \begin{pmatrix} \frac{\partial f_1}{\partial x} \\ \vdots \\ \frac{\partial f_m}{\partial x} \end{pmatrix} \cdot dx = y \cdot dx$$

$[m \times 1]$

Упражнение

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$[n \times 1]$

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nd} \end{bmatrix}$$

$[n \times d]$

$$w = \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix}$$

$[d \times 1]$

$$y = Xw + \varepsilon$$

$[n \times 1] \quad [n \times d] \quad [d \times 1] \quad [n \times 1]$

$$x_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{pmatrix}$$

$[d \times 1]$

$$a(x) = Xw$$

$[n \times 1] \quad [n \times d] \quad [d \times 1]$

$$\hat{y} = X\hat{w}$$

$$a(x_i) = x_i^T w = \langle x_i, w \rangle$$

$$L(w) = \frac{1}{n} \sum_{i=1}^n (y_i - a(x_i))^2 \rightarrow \min_w$$

MSE

$$\sum h_i^2$$

$h^T h$

$$y - a(x)$$

$$y - Xw$$

$$L(w) = \frac{1}{n} (y - Xw)^T (y - Xw) \rightarrow \min_w$$

$$L: \mathbb{R}^d \rightarrow \mathbb{R}$$

$$dL = \frac{1}{n} d(y - Xw)^T \cdot (y - Xw) + \frac{1}{n} (y - Xw)^T \cdot d(y - Xw) =$$

$$\begin{aligned} & d(-Xw)^T \\ & (-Xdw)^T \\ & - dw^T X^T \end{aligned}$$

$$= -\frac{1}{n} \cdot \left[\underbrace{dw^T}_{[1 \times d]} \underbrace{X^T}_{[d \times n]} \underbrace{(y - Xw)}_{[n \times 1]} + \underbrace{(y - Xw)^T}_{[1 \times n]} \underbrace{X}_{[n \times d]} dw \right] =$$

$$\left[dw^T X^T (y - Xw) \right]^T = (y - Xw)^T X dw$$

$$= \underbrace{-\frac{2}{n} (y - Xw)^T X}_{\nabla_w^T L} dw$$

$$\nabla_w L = -\frac{2}{n} X^T (y - Xw)$$

$$-\cancel{\frac{2}{n}} X^T (y - X\hat{w}) = 0$$

$$X^T y = X^T X \hat{w}$$

$$\hat{w} = (X^T X)^{-1} X^T y$$

А это только минимум?

$$\nabla_w^2 L$$

$$g(w) = -\frac{2}{n} X^T (y - Xw)$$

$$[d \times 1]$$

$$g: \mathbb{R}^d \rightarrow \mathbb{R}^d$$

$$dg = -\frac{2}{n} x^T (0 - x dw) = \frac{2}{n} x^T x dw$$

$y \cdot dw$

$$H = \nabla_w^2 L = 2 \cdot \frac{x^T x}{n}$$

полож. опред. $\Rightarrow \min$

отриц. опред. $\Rightarrow \max$

Опр.

H полож.

$$\forall \delta \in \mathbb{R}^n \setminus \{0\}$$

$$\delta^T H \delta > 0$$

$$\delta^T \cdot \frac{2}{n} \cdot x^T x \cdot \delta = \frac{2}{n} \cdot (x\delta)^T x\delta = \frac{2}{n} \tilde{\delta}^T \tilde{\delta} = \frac{2}{n} \sum \tilde{\delta}_i^2 > 0$$

$\Rightarrow$ λ в точке минимума

Упражнение

$$f(x) = x x^T x$$

$[n \times 1] \quad [n \times 1][1 \times n][n \times 1]$

$$\nabla_x f(x)$$

$$df = y \cdot dx$$

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

$$df = d(x x^T x) = dx \cdot x^T x + x \cdot d(x^T x) =$$

$$= dx \cdot x^T x + x \cdot dx^T \cdot x + x x^T dx =$$

$[n \times 1] \quad [1 \times 1]$
скаляр

$[1 \times 1]$

$[n \times n] \quad [n \times 1]$

$$[dx^T \cdot x]^T = x^T dx$$

$[1 \times 1]$

$$= x^T x \cdot dx + x x^T dx + x x^T dx =$$

$[1 \times 1] \quad [n \times 1]$

$$= \underset{\substack{[1 \times 1] \\ \uparrow \\ I_n}}{x^T x} \cdot dx + 2 \underset{\substack{[1 \times 1] \\ [n \times 1]}}{x} \underset{\substack{[n \times 1]}}{x^T} dx = \left(\underset{\substack{[1 \times 1] \\ [n \times n]}}{x^T x \cdot I_n} + 2 \underset{\substack{[1 \times 1] \\ [n \times n]}}{x x^T} \right) dx$$

избранский
фокус

$$L = x^T x \cdot I_n + 2 x x^T$$

$$4 \cdot \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} = \begin{pmatrix} 4 dx_1 \\ 4 dx_2 \end{pmatrix} = \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} \cdot 4$$

$$4 \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} = \begin{pmatrix} 4 \cdot dx_1 \\ 4 \cdot dx_2 \end{pmatrix}$$